
SAM GUESSOUM

AI RESEARCHER | MACHINE LEARNING | MEDICAL AI
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RESEARCH PROFILE

Artificial intelligence researcher completing an M.S. at Worcester Polytechnic Institute with a 4.0 GPA and transitioning to doctoral study in Computer Science. Research spans medical AI, computer vision, explainable decision support, reinforcement learning, and multi-agent optimization, with peer-reviewed work accepted across IEEE, Springer LNCS, and Computing in Cardiology venues.

EDUCATION

- M.S. Artificial Intelligence** | Worcester Polytechnic Institute Aug 2024 - Aug 2026
GPA: 4.0/4.0 | Thesis: Multi-Agent Optimization for 2D and 3D Bin Packing
- B.Sc. Applied Mathematics** | University of Greenwich Sep 2020 - Aug 2023
Machine Learning examination: 85% | Capstone: English Premier League match prediction

RESEARCH & PROFESSIONAL EXPERIENCE

- Incoming Research Intern** | Harvard Ophthalmology AI Lab, Massachusetts Eye and Ear / Harvard Medical School Sep 2026 - May 2027
 - Developing a retrieval-grounded system for cataract-surgery video analysis, combining temporal phase recognition, tool detection, structured timelines, and evidence-linked question answering.
 - Built an evaluation pipeline using ResNet18, MS-TCN, and YOLOv8; achieved 93.11% phase accuracy, 0.908 macro-F1, 0.807 segment IoU, and 11.4-second mean timestamp error on the locked test set.

- Research Assistant** | Worcester Polytechnic Institute Jan 2025 - Present
 - Investigate NP-hard 1D, 2D, and 3D bin-packing problems using multi-agent, heuristic, metaheuristic, and learning-based methods across public benchmark datasets.
 - Design reproducible experiments and evaluate solution quality, runtime, scalability, and generalization across problem sizes and packing geometries.

- Research Intern** | Findability Sciences Aug 2025 - Dec 2025
 - Built a multi-objective Bayesian optimization workflow for industrial sugarcane milling, reducing 150+ sensor variables into interpretable control features.
 - Produced Pareto-optimized operating recommendations associated with an approximately 4.8% simulated efficiency improvement; received WPI Data Science GQP MVP recognition.

- Research Intern** | thyssenkrupp Materials Services Jan 2025 - May 2026
 - Developed multi-agent approaches for rectangular bin packing under industrial constraints, contributing to a peer-reviewed paper and its 3D extension.

- Research Collaborator** | RISE Lab, Johns Hopkins University - Supervisor: Samuel Miles Mar 2026 - Present
 - Contribute to the DRC3B systems-analysis study of electricity reliability and medical-oxygen access across health facilities in three provinces of the Democratic Republic of the Congo.
 - Harmonize DRC2 survey and health-facility data with DRC3a high-frequency power and oxygen device-energy data for facility classification, outage and voltage analysis, and oxygen-system vulnerability assessment.
 - Support reproducible data pipelines, cross-province visualizations, predictive reliability modeling, energy-constrained oxygen planning, and manuscript deliverables.

- Graduate Student Ambassador** | Worcester Polytechnic Institute Feb 2025 - Present
 - Guide prospective and admitted graduate students through academic programs, onboarding, campus tours, and virtual information sessions while supporting outreach operations.

SELECTED PUBLICATIONS & RESEARCH

- A Multimodal Decision-Support Framework for Cardiology Triage with Learning-Based Test Selection and Explainable AI** — Computing in Cardiology 2026 (*accepted, regular track*)
- UPMO: An Uncertainty-Aware Predictive Multi-Objective Optimization Framework for Smart Agriculture** — 16th IEEE International Conference on Advanced Computer Information Technologies (ACIT 2026) (*accepted*)
- Towards Real-Time Precision Agriculture: A Task-Aware Multi-Modal Decision Framework** — IEEE International Conference on Intelligent Systems and Interdisciplinary Applications (ISAIA 2026) (*accepted; co-author*)
- NeuroGraphMamba** — Applications of Medical AI Workshop 2026 (*accepted; co-author*)
- Automated Surgical Report Generation via Retrieval-Augmented LLMs for Cataract Surgery Video Understanding** — SPIE Medical Imaging 2027 - Imaging Informatics (*manuscript in preparation; with Mohammad Eslami*)
- An Adaptive Multi-Agent System with Data-Driven Heuristic Orchestration for Solving 1D/2D Bin-Packing Problems** — peer-reviewed conference paper; Springer LNCS extended version (*accepted; with Chun-Kit Ngan and Rolf Bardeli*)
- Data-Driven Optimization of Sugarcane Milling Processes** — Worcester Polytechnic Institute Data Science GQP (*research poster, 2025*)

SELECTED TECHNICAL PROJECTS

- Cataract Surgery Video Intelligence** | Independent research prototype 2026
- Created an end-to-end pipeline for video upload, frame sampling, phase recognition, temporal smoothing, tool/event detection, grounded Q&A, uncertainty flags, and structured report export.
 - Evaluated cross-dataset transfer between Cataract-101 and Cataract-1K and documented calibration, ablations, runtime, retrieval examples, and failure cases.
- Cardiology Triage Decision Support** | Research system 2026
- Designed an ECG-centered multimodal framework with explainable AI and adaptive diagnostic-test recommendation; evaluated on approximately 500 cardiologist-annotated ECGs and public benchmarks including MIT-BIH and PTB-XL.
- Brain-Computer Interface Surgical Guidance** | Decision-support application 2025
- Engineered spatiotemporal feature extraction for 1,024-channel micro-ECoG streams sampled at 500 Hz and built an interactive GO/NO-GO cortical-placement guidance interface.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL, HTML/CSS, R

Machine Learning: PyTorch, scikit-learn, pandas, NumPy, computer vision, deep learning, reinforcement learning, statistical modeling, explainable AI

Methods: ResNet, MS-TCN, YOLO, random forests, SVM, KNN, PCA, Bayesian optimization, genetic algorithms, ant colony optimization, heuristic search

Research & Tools: Experimental design, cross-validation, ablation studies, error analysis, Git, Streamlit, FastAPI, LaTeX, data visualization

Languages: Arabic, English, and French (fluent); Spanish (intermediate)

SELECTED RESEARCH TRAINING

Introduction to Computational Neuroscience Summer School — Arabs in Neuroscience | Jul 13-31, 2026

5th Surgical Data Science Summer School (online) — IHU Strasbourg / University of Strasbourg | Jun 8-12, 2026

Connecting Your IoT Devices (96 hours, 4 credits; hybrid) — European Digital UniverCity, ENSSAT / University of Rennes | Jun 10-30, 2026

HONORS, LEADERSHIP & SERVICE

- First Place, Tech for Mobility: Transforming Sidi Abdellah's Transport Future, University of Algiers 1 (2025)
- First Place, AfriDev Hackathon, Worcester Polytechnic Institute (2025)
- Second Place, Hacking the Racial Wealth Gap: Jobs & Entrepreneurship, MIT Community Innovators Lab and partners (2025)
- Most Valuable Player, WPI Data Science Graduate Qualifying Project (2025)
- Peer Reviewer, AFRICAI Workshop at MICCAI 2026
- University of Greenwich Soccer Team; Mathematics Club